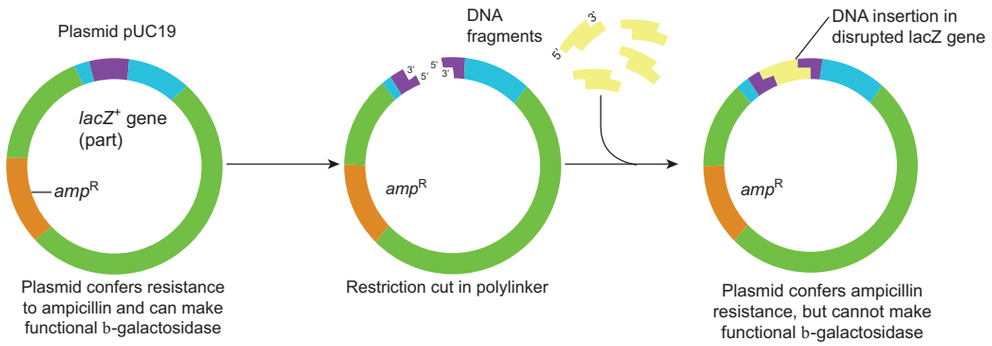


Figure 12.25: Insertion of a piece of DNA into the plasmid cloning vector pUC19 to produce a recombinant DNA molecule

4. The MCS is inserted into part of the E.coli β – galactosidase (lac Z^+) gene. Figure 12.25 illustrates how a piece of DNA can be inserted into a plasmid cloning vector such as pUC19

Bacteriophage as Cloning Vectors

They are viruses that replicate within the bacteria. A phage can be employed as vector since a foreign DNA can be spliced into phage DNA, without causing harm to phage genes. The phage will reproduce (replicate the foreign DNA) when it infects bacterial cell. Both single and double stranded phage vectors have been employed in recombinant DNA technology. Derivatives of phage can carry fragments up to about 45 kb in length. PI bacteriophage can carry fragments up to 95 kb

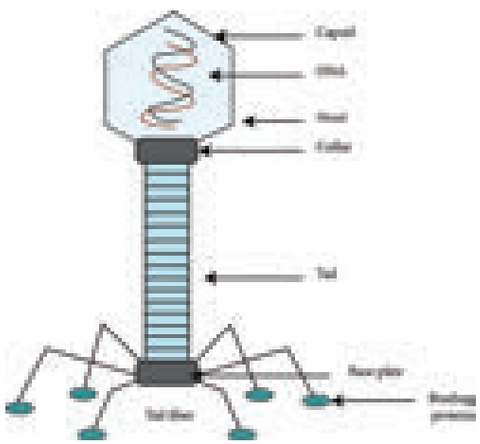


Figure 12.26: Structure of phage λ

Phage λ (Figure 12.26) consists of a head and a tail (both proteins). The DNA, located in the head, is a linear molecule of about 50 kb. At each end of the DNA, there are single stranded extensions called

Infobits

Terms associated with plasmids:

1. Low copy number plasmids are plasmids that occur low in number in each cell.
2. High copy number are plasmids that occur high in number in each cell.
3. Conjugative plasmids carry a set of transfer genes (tra genes) that facilities bacterial conjugation.
4. Non – conjugative plasmidsare plasmids that do not possess transfer genes.
5. Stringent plasmids are plasmids that are present in a limited number (1–2 per cell).
6. Relaxed plasmids are plasmids that occur in large number in each cell.
7. F plasmids possess genes for their own transfer from one cell to another
8. R plasmids carry genes resistance to antibiotics.

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